

Ejercicio

martes, 14 de octubre de 2025

12:49 p. m.

$$(1 - 0.388B^4) \nabla T(CETES) = (1 - 0.481B^2) \alpha_t$$

con $\hat{\sigma}_\alpha = 1104.7$ y $T(CETES) = \frac{CETES_t}{1000}$

$$T(CETES) = X_t \quad \nabla X_t = (1 - B)X_t$$

$$(1 - 0.388B^4) (1 - B)X_t = (1 - 0.481B^2) \alpha_t$$

$$(1 - 0.388B^4) (X_t - X_{t-1}) = (1 - 0.481B^2) \alpha_t$$

$$X_t - 0.388X_{t-4} - X_{t-1} + 0.388X_{t-5} = \alpha_t - 0.481\alpha_{t-2}$$

$$X_t = X_{t-1} + 0.388X_{t-4} - 0.388X_{t-5} + \alpha_t - 0.481\alpha_{t-2}$$

Se toma $\hat{X}_{t+h} = E[X_{t+h}]$

Para $h=1$

$$\hat{X}_{t+1} = E[X_{t+1}]$$

Dado que conocemos

$$X_t = X_{t-1} + 0.388X_{t-4} - 0.388X_{t-5} + \alpha_t - 0.481\alpha_{t-2}$$

$$X_{t+1} = X_t + 0.388X_{t-3} - 0.388X_{t-4} + \alpha_{t+1} - 0.481\alpha_{t-1}$$

$$E[X_{t+1}] = E[X_t] + 0.388E[X_{t-3}] - 0.388E[X_{t-4}] + E[\alpha_{t+1}] - 0.481E[\alpha_{t-1}]$$

- $E[X_t] \quad h-j=0 \quad h \neq j$

$$X_t = T_{54} = 7561.768$$

- $E[X_{t-3}] \quad h-j=-3 \quad h < j$

$$X_{t-3} = T_{56} = 8661.59$$

- $E[X_{t-4}] \quad h-j=-4 \quad h < j$

$$X_{t-4} = T_{55} = 10023.351$$

- $E[\alpha_{t+1}] = 0$

- $E[\alpha_{t-1}] = \hat{\alpha}_{t-1} = \alpha_{58} = (-1379.3)$

$$\Rightarrow E[X_{t+1}] = T_{54} + 0.388T_{56} - T_{55} + 0 - 0.481\alpha_{58}$$

$$\Rightarrow E[X_{t+1}] = T_{54} + 0.388 T_{56} - T_{55} + 0 - 0.481 \alpha_{56}$$

$$\hat{X}_{t+1} = 7696.848$$

Para $n=2$

$$\hat{X}_{t+2} = E[X_{t+2}]$$

$$E[X_{t+2}] = E[X_{t+1}] + 0.388 E[X_{t-2}] - 0.388 E[X_{t-3}] + E[\alpha_{t+2}] - 0.481 E[\alpha_t]$$

$$\bullet E[X_{t+1}] \quad n > j \quad n-j=1$$

$$\hat{X}_{t+1} = 7696.848$$

$$\bullet E[X_{t-2}] \quad n < j \quad n-j=-2$$

$$X_{t-2} = T_{57} = 8612.446$$

$$\bullet E[X_{t-3}] \quad n < j \quad n-j=-3$$

$$X_{t-3} = T_{56} = 8661.59$$

$$\bullet E[\alpha_{t+2}] = 0$$

$$\bullet E[\alpha_t] = \alpha_t = -548.433$$

$$\hat{X}_{t+2} = \hat{X}_{t+1} + 0.388 T_{57} - 0.388 T_{56} + 0 - 0.481(-548.433)$$

$$\hat{X}_{t+2} = 7941.575$$

Para $n=3$

$$\hat{X}_{t+3} = E[X_{t+3}]$$

$$= E[X_{t+2}] + 0.388 E[X_{t-1}] - 0.388 E[X_{t-2}] + E[\alpha_{t+3}] - 0.481 E[\alpha_{t+1}]$$

$$\bullet E[X_{t+2}] = \hat{X}_{t+2} = 7941.575$$

$$\bullet E[X_{t-1}] = X_{t-1} = T_{58} = 8015.310$$

$$\bullet E[X_{t-2}] = X_{t-2} = T_{57} = 8612.446$$

$$\bullet E[\alpha_{t+3}] = 0$$

$$-E[\alpha_{t+1}] = 0$$

$$\hat{X}_{t+3} = \hat{X}_{t+2} + 0.388 T_{58} - 0.388 T_{57}$$

$$= 7709.886$$

INTERVALOS

$$\phi_1 = 1 \quad \phi_2 = 0 \quad \phi_3 = 0 \quad \phi_4 = 0.388 \quad \phi_5 = -0.388$$

$$\theta_1 = 0 \quad \theta_2 = -0.481$$

$$\hat{\psi}_0 = 1$$

$$\hat{\psi}_1 = 0$$

$$\hat{\psi}_2 = 0.481$$

Intervalos de pred:

$$z_t(h) \pm z_{\frac{\alpha}{2}} \hat{\sigma}_\alpha \sqrt{\sum_{j=0}^{h-1} \psi_j^2}$$

para intervalo 1.96

para $h=1$

$$\hat{X}_{t+1} \pm 1.96 (1109.7) ((-1)^2)^{1/2}$$

$$(5521.88, 9811.816)$$

para $h=2$

$$\hat{X}_{t+2} \pm 1.96 (1109.7) (1)$$

$$(5766.607, 11017.5065)$$

para $h=3$

$$\hat{X}_{t+3} \pm 1.96 (1109.7) (\sqrt{-1^2 + 0.481^2})$$

$$\hat{X}_{++3} \pm 1.96 (1109.7) (\sqrt{-1^2 + 0.481^2})$$

$$(5296.4, 10123.4)$$